DIVYA RAUT

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Scientist with hands-on experience building end-to-end machine learning pipelines using Python, XGBoost, and Scikit-learn. Skilled in predictive modeling, survival analysis, and deploying production-ready APIs using FastAPI and Docker. Experienced in delivering actionable insights from complex datasets across telecom, healthcare, and network security domains.

TECHNICAL SKILLS

API & Backend: FastAPI (REST API Design, GET/POST Methods), Python, MYSQL
Deployment & Ops: Docker, Docker Compose, Render, Git/GitHub, LangChain (learning)
Machine Learning: XGBoost, Random Forest, Scikit-learn, Feature Engineering, Model Evaluation (ROCAUC, F1-Score), Linear Regression, Logistic Regression
Data & Visualization: Pandas, NumPy, Power BI, Matplotlib

PROFESSIONAL EXPERIENCE

Machine Learning Developer | Machine Learning Engineer Projects
Designed and developed end-to-end Machine Learning pipelines, moving from raw data to deployed REST APIs.
Built and documented FastAPI microservices to serve model predictions in real-time using efficient POST requests.
Containerized full-stack applications using Docker, ensuring 100% consistency between development and production environments.
Developed client-facing Streamlit applications to demonstrate model capabilities and allow non-technical users to interact with data.
Conducted rigorous model validation and optimization, handling class imbalance and tuning hyperparameters for maximum accuracy.

PROJECTS

Real-Time Intrusion Detection System (FastAPI + Docker) | https://hub.docker.com/r/dataforai/intrusion-detection-api
Objective: Built a system to classify network traffic as "Normal" or "Attack" in real-time.
Tech Stack: Python, Scikit-learn, FastAPI, Docker.
Implementation: Trained an ensemble classifier on high-dimensional data and exposed it via a FastAPI POST endpoint to return threat probabilities reliably.

Medical Insurance Risk Prediction API | https://hub.docker.com/r/dataforai/insurance-risk-api
Objective: Predict high-risk insurance applicants to optimize premium pricing.
Tech Stack: Python, Logistic Regression, Streamlit, Docker.
Implementation: Developed a predictive model with a Streamlit frontend, allowing users to input metrics and see risk scores instantly via a backend API.

Telecom Customer Churn Analysis| https://hub.docker.com/r/dataforai/churn-prediction-api
Objective: Identify customers likely to cancel their service to improve retention.
Tech Stack: XGBoost, Pandas, Survival Analysis.
Implementation: Engineered features and handled class imbalance using scale_pos_weight. Achieved 0.85 ROC-AUC, enabling identification of high-risk customers to prioritize retention efforts and reduce revenue loss

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EDUCATION : Bachelor of Computer Applications (BCA)